— Freshman year —

Physics research in UIUC Professor Alexey Bezryadin’s Lab:

Joined Professor Alexey Bezryadin’s Lab.

Developed critical current model for generalized superconducting quantum interference devices (SQUIDs) with tunnel Josephson Junctions (JJs)

Published python package used to generate random numbers corresponding to specific standard deviation, used to model effects of fabrication disorders in JJ-SQUIDs

Proved rotational symmetries of JJ-SQUIDs, typed up report and presented results at the annual UIUC Undergraduate Research Symposium.

Solved decade-long “node-lifting” problem using my JJ-SQUID critical current model.

Revived a decade-long broken time-resolved circuit-QED measurement set-up, used it to reproduce non-linear phenomenon in cavity-coupled nanowires

NASA competition:

Contributed to systems-level design of a lunar regolith density sensor. Awarded $9k, awarded 2nd place, and design was incorporated into Artemis mission architecture.

— Summer 2024 —

Physics research in UIUC Professor Alexey Bezryadin’s Lab:

Developed numerical optimization tool using Python and C++ to reconstruct JJ-SQUID geometry from experimental data, using gradient-based optimization and parallel processing. Tool now used by other UIUC research groups.

— Sophomore year —

Physics research in UIUC Professor Alexey Bezryadin’s Lab:

Built and investigated critical current model of many-nanowire SQUIDs, incorporating in metastability property. Discovered global device symmetry, used group theory to prove isomorphism with particle physics. Published first paper (first-author) on metastable investigation and symmetry isomorphism.

Developed group theoretical explanation of nanowire diode behavior through parity-time symmetry breaking, leveraging previously established isomorphism. Published results as second paper (first-author).

NASA competition:

Recruited and led a 40+ member NASA competition team to design first hardware-aware cryogenic propellant transfer architecture.

Built two-phase flow regime classification algorithm and hardware-aware line chilldown protocol.

Awarded $9k funding, presented at NASA, and ideas were integrated into Artemis mission architecture.

— Summer 2025 —

Research internship at Johns Hopkins Applied Physics Lab:

Derived and implemented holographic generative quantum machine learning algorithm.

Developed a quantum-annealing approach for radar optimization, becoming the first APL intern to get a project funded.

Rehired for the semester.

— Junior year —

Physics research in UIUC Professor Alexey Bezryadin’s Lab

Discovered first nanowire transmon-qubit concept by building and verifying a phenomenological theory of nanowires at low temperatures. Solved a 20-year problem that the group had been actively pursuing. Published results as third paper (first-author) and filed a U.S. patent

My group theoretical symmetry breaking prediction was experimentally verified, resulting in a fourth paper (second-author).

Awarded 2026 Barry M. Goldwater Scholarship and 2026 Astronaut Scholarship.

Research internship at Johns Hopkins Applied Physics Lab:

Explored machine learning applications to learn and generalize quantum circuit compilations from classical state representations. Built and validated a transformer model with various positional encodings schemes. Built tokenization scheme native to quantum assembly code (openQASM).

— Summer 2026 —

Research internship at Johns Hopkins Applied Physics Lab:

Designed a hybrid machine learning architecture (Large Language Model and Long Short-Term Memory Model) to learn and generalize holographic quantum circuits. Benchmarked on the generalized W state, doubling size of classical register with 99% accuracy. Wrote a fifth paper (first-author), now currently under internal review before release.

Built matrix-product-states solver for dynamical simulations of nonlinear diffusion-advection PDEs. Demonstrated classical scaling advantage for high grid points and designed bond dimension-efficient technique for taking functions of tensor networks. Now preparing results in the sixth paper (first-author).

Developed and validated semi-classical cold atom model for state estimation. Released as a first-author internal memorandum.

— Senior year —

Physics research in UIUC Professor Alexey Bezryadin’s Lab

Built theoretical framework for quantum transduction based on interactions between optical photons and superconducting vortices. Now preparing results as seventh paper (first-author)

Research internship at Johns Hopkins Applied Physics Lab:

Currently applying hybrid machine learning architecture to scale up expressiveness (ancilla qubit count) of quantum materials.

Architected multi-layer perceptron using tensor networks, using my previously developed bond dimension-efficient technique for taking functions of tensor networks. Preparing results in an eighth paper (first-author).